

CUXHAVEN NORDHOLZ, Germany

WMO: 101360

Lat: 53.7647N

Lon: 8.6583E

Elev: 32

StdP: 100.94

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.3	-6.3	-11.0	1.5	-7.0	-8.9	1.8	-5.7	15.1	6.7	13.5	6.7	3.6	100	0.618

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.1	28.6	19.4	26.1	18.6	24.0	17.7	20.4	26.4	19.4	24.8	18.4	22.9	4.2	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.2	22.4	17.3	12.4	21.5	16.5	11.8	20.7	58.5	26.3	55.3	24.8	52.1	22.8	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.9	10.4	9.2	-10.8	32.4	3.1	1.6	-13.1	33.6	-14.9	34.5	-16.7	35.5	-18.9	36.7		
			-11.1	22.6	3.0	1.0	-13.3	23.4	-15.1	24.0	-16.8	24.6	-19.0	25.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	1.8	2.3	4.6	8.7	12.4	15.3	17.8	17.8	14.6	10.3	5.9	2.8
	DBStd	6.72	4.26	3.99	3.52	3.57	3.39	3.03	3.04	2.73	2.62	3.27	3.42	4.20
	HDD10.0	1107	254	216	169	67	13	0	0	0	1	36	129	222
	HDD18.3	3276	512	449	426	289	187	102	48	44	117	248	374	480
	CDD10.0	947	0	1	2	28	87	158	242	241	137	46	4	1
	CDD18.3	76	0	0	0	0	3	10	32	27	4	0	0	0
	CDH23.3	611	0	0	0	3	33	91	265	193	26	0	0	0
	CDH26.7	160	0	0	0	0	4	22	77	55	3	0	0	0
Wind	WSAvg	4.5	5.1	5.0	5.1	4.5	4.4	4.3	4.1	3.9	4.0	4.3	4.4	4.8
Precipitation	PrecAvg	853	66	52	50	40	57	84	92	91	85	82	79	75
	PrecMax	1140	140	116	93	97	116	182	182	216	200	201	164	189
	PrecMin	499	1	18	9	6	12	7	39	27	27	27	10	17
	PrecStd	141	34	26	25	20	25	43	43	46	46	44	38	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	12.1	16.8	23.0	26.8	29.3	31.9	31.0	26.2	20.2	15.1	12.2
		MCWB	9.4	8.4	10.7	14.1	17.8	19.6	20.5	19.9	18.7	15.9	13.0	10.6
	2%	DB	9.7	10.2	13.2	20.1	23.8	26.0	28.9	28.1	23.2	17.9	13.1	10.9
		MCWB	8.4	8.0	9.2	12.6	15.8	18.3	19.6	19.2	17.6	14.9	11.6	9.6
	5%	DB	8.2	8.9	11.1	17.6	21.2	23.1	26.2	25.1	20.8	16.1	11.8	9.2
		MCWB	7.2	7.4	8.4	11.8	14.7	17.1	18.9	18.6	16.3	13.8	10.3	8.2
	10%	DB	7.1	7.6	9.5	15.0	18.8	20.8	23.9	22.9	18.9	14.9	10.5	8.1
		MCWB	6.2	6.3	7.5	10.7	13.6	15.9	18.0	17.6	15.3	13.0	9.2	7.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	9.8	11.7	15.1	19.2	21.0	21.8	21.8	19.6	16.4	13.5	11.2
		MCDB	11.0	11.0	14.9	21.1	25.0	27.4	28.5	27.6	24.7	19.4	14.7	12.1
	2%	WB	8.5	8.3	10.0	13.5	16.9	19.2	20.5	20.1	18.0	15.1	11.8	9.8
		MCDB	9.4	9.6	12.3	18.5	21.8	24.7	26.9	25.6	22.2	17.3	13.0	10.6
	5%	WB	7.2	7.3	8.8	12.3	15.2	17.7	19.5	19.1	16.7	14.1	10.5	8.4
		MCDB	8.0	8.6	10.8	16.5	19.7	21.8	25.2	23.8	20.0	15.8	11.6	9.2
	10%	WB	6.2	6.2	7.5	11.0	14.0	16.5	18.4	18.2	15.7	13.2	9.4	7.2
		MCDB	6.9	7.3	9.2	14.6	18.1	20.0	23.0	22.3	18.1	14.7	10.3	7.9
Mean Daily Temperature Range	5% DB	MDBR	4.1	4.9	6.3	8.4	8.5	7.9	8.1	8.1	7.2	6.1	4.5	4.1
		MCDBR	4.7	5.9	9.3	13.0	13.1	12.4	12.8	12.2	10.5	7.3	4.9	4.5
		MCWBR	4.2	4.6	6.1	7.0	6.9	6.5	5.7	5.7	5.9	5.0	4.1	4.1
	5% WB	MCDBR	4.4	5.1	8.2	11.1	11.5	11.1	11.4	10.4	9.3	6.8	4.7	4.4
		MCWBR	4.1	4.3	5.8	6.6	6.7	6.5	5.6	5.5	5.6	5.0	4.2	4.2
Clear-Sky Solar Irradiance	taub	0.326	0.345	0.374	0.407	0.401	0.400	0.416	0.410	0.386	0.362	0.339	0.318	
	taud	2.313	2.329	2.270	2.235	2.300	2.327	2.295	2.321	2.375	2.399	2.350	2.306	
	Ebn at Noon	626	743	798	819	847	850	827	808	781	715	605	556	
	Edh at Noon	60	80	104	123	121	119	121	112	94	74	58	50	
All-Sky Solar Radiation	RadAvg	0.56	1.22	2.43	4.03	5.06	5.27	5.02	4.14	2.98	1.65	0.70	0.42	
	RadStd	0.06	0.17	0.24	0.47	0.54	0.43	0.65	0.31	0.35	0.18	0.07	0.03	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-132	N/A	N/A

Nomenclature: See separate page